



Fidji Berio

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 fidji.berio@gmail.com

 French Citizenship

Social Networks

 Bluesky

 ResearchGate

 Google Scholar

 Github

 Orcid

 LinkedIn

Languages

 French ● ● ● ● ●

 English ● ● ● ● ●

 Spanish ● ● ● ● ●

Certifications

 Level 2 FFESSM (Advanced PADI)

 Level 1 Nitrox

 PADI Freediver

Hard Skills

 R programming (10 years)

 MicroCT imaging (7 years, >300 scans)

 Geometric morphometrics (5 years)

 Machine learning (2 years)

 microCT scan segmentation (3 years)

 High-speed videos (2 years)

 Histology (2 years)

 Biomechanics (2 years)

 Functional tests (4 years)

 Literature review (7 years)

PhD in Biology

Evo-Devo – Climate change – Biomechanics – Chondrichthyans

Working Experience

Jan 2025–
Mar 2026

Postdoctoral researcher

Scripps Research Institute, USA

Projects Locomotion biomechanics in fishes & Skeletal mineralization in fish under climate change [1–3, 5–7]

Methods 3D analysis of high-speed videos of swimming and walking selachians and actinopterygians, microCT scan segmentation of actinopterygians, bone density analyses, wavelet analyses on otoliths outlines, respirometry analyses

Nov 2022–Dec
2024

Postdoctoral researcher

Stockholm University, Sweden

Projects Fish biomechanics [4, 8–9, 11]

Methods 3D analysis of high-speed videos of swimming and walking selachians and actinopterygians, microCT scan acquisitions and segmentations (selachians and actinopterygians from museums), respirometry experiments in teleosts, fluorescence staining in selachians, fluorescence microscopy, dissections of teleosts, student supervision, project management, fish care

Sept–Oct 2022

Guest professor

University of Vienna, Austria

Teaching Functional evolution of jaw structures

Méthodes colorations iodine et microCT scans de chimères, enseignement

2021–2022

Science communicator

Futura Sciences

Activities Online writing of [popularization articles](#), interviews

Topics Marine biology, paleontology, paleoanthropology

2017 – 2021

PhD

ENS Lyon & Institute of Evolution Sciences of Montpellier, France

Student

Main project Physical and genetic factors impacting development and evolution of elasmobranch odontodes [10, 12–15, 18–21]

Methods microCT imaging of shark teeth, histology of chondrichthyan teeth and vertebrae, machine learning and geometric morphometrics with R on selachian teeth, functional tests on *Scyliorhinus canicula* embryos, histology and comparative anatomy of vertebrae, dissections of selachian jaws, study of a chimaera (*Hydrolagus collieri*) reproduction with the aquarium of Montpellier, student supervision, literature review, project management

2017

Master Thesis

Univ. du Québec à Rimouski, Canada

Project Regionalisation of the vertebral column in a skate (*Amblyraja radiata*) [17]

Methods microCT imaging, character matrices, clustering with R, Clear & Stain double coloration, drawing with a camera lucida

2016

Master Thesis

Oceanological Observatory of Banyuls-sur-mer, France

Project Long-term experiment on growth dynamics of hydrothermal mussels *Bathymodiolus thermophilus* related to hydrothermal fluids dynamics

Methods Sclerochronology, Fast Fourier Transforms, Mutvei staining

2015

Bachelor Thesis

Institute for Cognitive and Integrative Neuroscience, France

Project Effects of serotonin on aggressiveness of the invasive Louisiana crayfish *Procambarus clarkii*

Methods Video tracking with EthoVision, ethology, functional tests on mature specimens, interaction videos, statistics with R

Education

2017 – 2021	PhD Studies Evo-devo of chondrichthyans	ENS Lyon & Institute of Evolution Sciences of Montpellier
2015 – 2017	Master Studies (1/31) Littoral and Sea Sciences with focus on Marine ecosystems and Polar environments	European Institute for Marine Studies, Brest
2012 – 2015	Bachelor Studies (1/60) Biology of Organisms and Ecosystems	Univ. of Bordeaux
2010 – 2012	Bachelor Studies Hypokhâgne & Khâgne – Literature studies (French and Spanish)	Univ. of Montpellier

Teaching

2018 – 2024	Bachelor & Master Students Advisor	ENS Lyon, Univ. of Montpellier, Stockholm Univ.
2022	To Master & PhD Students Functional Evolution of Jaw Structures	Univ. of Vienna
2019	To Bachelor Students Integrative Biology of Organisms and Comparative Anatomy	Univ. of Montpellier
2019	To High School Students Embryo and Evolution	Univ. of Montpellier
2018	To Bachelor Students Life Cycles – Genetics	Univ. of Montpellier

Conferences

2023	Poster presentation Society for Experimental Biology	United Kingdom
2022	Oral presentation Sharks International	Online Edition
2020	Oral presentation ToScA Global	Online Edition
2019	Oral presentation Comparative Cartilage Biology meeting	France
2018	Poster presentation Evolution	France
2018	Poster presentation EuroEvoDevo	Ireland
2018	Oral presentation Symposium of Morphometry and Evolution of Forms	France

Referees



Valentina Di Santo, Assistant professor, Scripps Research Institute – Postdoc advisor
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Tunhe Zhou, Research Engineer, Stockholm University – Microtomography
Department
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Mélanie Debiais-Thibaud, Professor, University of Montpellier – PhD advisor
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Richard Cloutier, Emeritus professor, Université du Québec à Rimouski – Master's
thesis advisor
richard_cloutier@uqar.ca

Publications

In prep

- [1] F. Berio and D. S. Valentina. “Century-scale changes in skeletal mineralization of a marine fish under climate change”. In: *Biological Journal of the Linnean Society* (2026).

Online

- [2] N. Khadem Hosseini, M. Ishida, F. **Berio**, V. Di Santo, and F. Iida. “A minimalistic walking fish robot twin based on the single actuator wave-like mechanism”. In: *Bioinspiration & Biomimetics* (2026). DOI: [10.1088/1748-3190/ae1fc8](https://doi.org/10.1088/1748-3190/ae1fc8).
- [3] Y. Gong, R. Sterling, X. Qi, F. **Berio**, O. Akanyeti, and V. Di Santo. “Running on empty: Locomotor compensation preserves fish schooling under hypoxia and informs principles for bioinspired swarms”. In: *Bioinspiration & Biomimetics* (2025). DOI: [10.1088/1748-3190/ae17fd](https://doi.org/10.1088/1748-3190/ae17fd).
- [4] J. H. Gayford, K. Soares, and F. **Berio**. “Sexual ornamentation and weapons of sexual conflict in cartilaginous fishes”. In: *Reviews in Fish Biology and Fisheries* (2025). DOI: [10.1007/s11160-025-10000-9](https://doi.org/10.1007/s11160-025-10000-9).
- [5] V. Di Santo, X. Qi, F. **Berio**, A. Albi, and O. Akanyeti. “Inherent instability leads to high costs of hovering in near-neutrally buoyant fishes”. In: *Proceedings of the National Academy of Sciences* (2025). DOI: [10.1073/pnas.2420015122](https://doi.org/10.1073/pnas.2420015122).
- [6] S. Z. Marketaki, F. **Berio**, and V. Di Santo. “Compensatory sensory mechanisms in naïve blind cavefish navigating novel environments after lateral line ablation”. In: *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology* (2025). DOI: [10.1016/j.cbpa.2025.111863](https://doi.org/10.1016/j.cbpa.2025.111863).
- [7] F. **Berio**, C. Morerod, and V. Di Santo. “Speed-dependent locomotor patterns during steady swimming in a demersal shark”. In: *Journal of Fish Biology* (2025). DOI: [10.1111/jfb.70043](https://doi.org/10.1111/jfb.70043).
- [8] F. A. López-Romero, E. Villalobos-Segura, J. Türtcher, F. **Berio**, S. Stumpf, R. P. Dearden, J. Kriwet, and E. Maldonado. “Evolution of the Batoidea Pectoral Fin Skeleton: Convergence, Modularity, and Integration Driving Disparity Trends”. In: *Evolutionary Ecology* 39 (2025), pp. 111–134. DOI: [10.1007/s10682-025-10330-x](https://doi.org/10.1007/s10682-025-10330-x).
- [9] M. Ishida, F. **Berio**, V. D. Santo, N. H. Shubin, and F. Iida. “Paleo-inspired robotics as an experimental approach to the history of life”. In: *Science Robotics* 9.95 (2024), eadn1125. DOI: [10.1126/scirobotics.adn1125](https://doi.org/10.1126/scirobotics.adn1125).
- [10] O. J. Atake, F. **Berio**, M. Debais-Thibaud, and B. F. Eames. “Extant cartilaginous fishes share trabecular and areolar mineralization patterns, but not tesserae, and evidence for a paedomorphic chimaera skeleton”. In: *eLife* (2025). DOI: [10.7554/eLife.94900.1](https://doi.org/10.7554/eLife.94900.1).
- [11] F. **Berio**, C. Morerod, X. Qi, and V. Di Santo. “Ontogenetic Plasticity in Shoaling Behavior in a Forage Fish under Warming”. In: *Integrative And Comparative Biology* 63.3 (2023), pp. 730–741. DOI: [10.1093/icb/icad043](https://doi.org/10.1093/icb/icad043).
- [12] F. **Berio**, A. Éon, R. Charron, E. Meunier, J. Marie, D. Florent, M. Simonet, N. Verschraegen, and N. Hirel. “Husbandry conditions of spotted ratfish (*Hydrolagus colliei*, Chimaeriformes) in aquaria for successful embryonic development and long-term survival of juveniles”. In: *Zoo Biology* 43.2 (2023), pp. 188–198. DOI: [10.1002/zoo.21813](https://doi.org/10.1002/zoo.21813).
- [13] R. Zimm, F. **Berio**, M. Debais-Thibaud, and N. Goudemand. “A shark-inspired general model of tooth morphogenesis unveils developmental asymmetries in phenotype transitions”. In: *Proceedings of the National Academy of Sciences* 120.15 (2023), e2216959120. DOI: [10.1073/pnas.2216959120](https://doi.org/10.1073/pnas.2216959120).
- [14] F. **Berio**, Y. Bayle, S. Agret, D. Baum, and N. Goudemand. “3D models related to the publication: Hide and seek shark teeth in Random Forests: Machine learning applied to *Scyliorhinus canicula*”. In: *MorphoMuseum* (2020), pp. 3–6. DOI: [10.18563/journal.m3.164](https://doi.org/10.18563/journal.m3.164).
- [15] F. **Berio**, Y. Bayle, D. Baum, N. Goudemand, and M. Debais-Thibaud. “Hide and seek shark teeth in Random Forests: Machine learning applied to *Scyliorhinus canicula*”. In: *PeerJ* 10 (2022), e13575. DOI: [10.7717/peerj.13575](https://doi.org/10.7717/peerj.13575).
- [16] F. A. López-Romero, F. **Berio**, D. Abed-Navandi, and J. Kriwet. “Early shape divergence of developmental trajectories in the jaw of galeomorph sharks”. In: *Frontiers in Zoology* 19 (2022), p. 7. DOI: [10.1186/s12983-022-00452-1](https://doi.org/10.1186/s12983-022-00452-1).
- [17] F. **Berio**, Y. Bayle, C. Riley, O. Larouche, and R. Cloutier. “Phenotypic regionalization of the vertebral column in the thorny skate *Amblyraja radiata*: Stability and variation”. In: *Journal of Anatomy* 240.2 (2021), pp. 253–267. DOI: [10.1111/joa.13551](https://doi.org/10.1111/joa.13551).
- [18] F. **Berio**, M. Broyon, S. Enault, N. Pirot, F. A. López-Romero, and M. Debais-Thibaud. “Diversity and evolution of mineralised skeletal tissues in chondrichthyans”. In: *Frontiers in Ecology and Evolution* 9 (2021), p. 660767. DOI: [10.3389/fevo.2021.660767](https://doi.org/10.3389/fevo.2021.660767).
- [19] F. **Berio**. “Multiscale variation of 3D tooth forms in selachians and developmental and evolutionary inferences: Odyssey of a scyliorhinid tooth”. PhD thesis. Institut de Génomique Fonctionnelle de Lyon and Institut des Sciences de l’Evolution de Montpellier, 2021. URL: <https://theses.hal.science/tel-03279588>.
- [20] F. **Berio**, A. Evin, N. Goudemand, and M. Debais-Thibaud. “The intraspecific diversity of tooth morphology in the large-spotted catshark *Scyliorhinus stellaris*: insights into the ontogenetic cues driving sexual dimorphism”. In: *Journal of Anatomy* 237 (2020), pp. 960–978. DOI: [10.1111/joa.13257](https://doi.org/10.1111/joa.13257).
- [21] F. **Berio** and M. Debais-Thibaud. “Evolutionary developmental genetics of teeth and odontodes in jawed vertebrates: a perspective from the study of elasmobranchs”. In: *Journal of Fish Biology* 98.4 (2019), pp. 906–918. DOI: [10.1111/jfb.14225](https://doi.org/10.1111/jfb.14225).